

Модуль 4. Лабораторная 3

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN.

Настройка PC:

```
PC4> ip 192.168.1.2/24
PC5> ip 192.168.1.3/24
PC6> ip 192.168.1.4/24
PC4> ip 192.168.1.5/24
PC5> ip 192.168.1.6/24
PC6> ip 192.168.1.7/24
```

Создание VLAN на всех коммутаторах:

```
enable
configure terminal
vlan 20
  name VLAN20
exit
vlan 333
  name VLAN333
exit
```

Layer2Switch-1:

```
interface range GigabitEthernet0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit
```

```
interface range GigabitEthernet1/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit
```

```
end
```

Layer2Switch-2:

```
enable
configure terminal

interface range GigabitEthernet0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit
```

```
interface range GigabitEthernet1/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit

end
```

Layer2Switch-3:

```
enable
configure terminal
interface range GigabitEthernet0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit

interface GigabitEthernet1/0
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface GigabitEthernet1/1
  switchport mode access
  switchport access vlan 333
  no shutdown
exit
```

Layer2Switch-4:

```
enable
configure terminal
interface range GigabitEthernet0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit

interface GigabitEthernet1/0
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface GigabitEthernet1/1
  switchport mode access
  switchport access vlan 333
  no shutdown
exit
```

Layer2Switch-5:

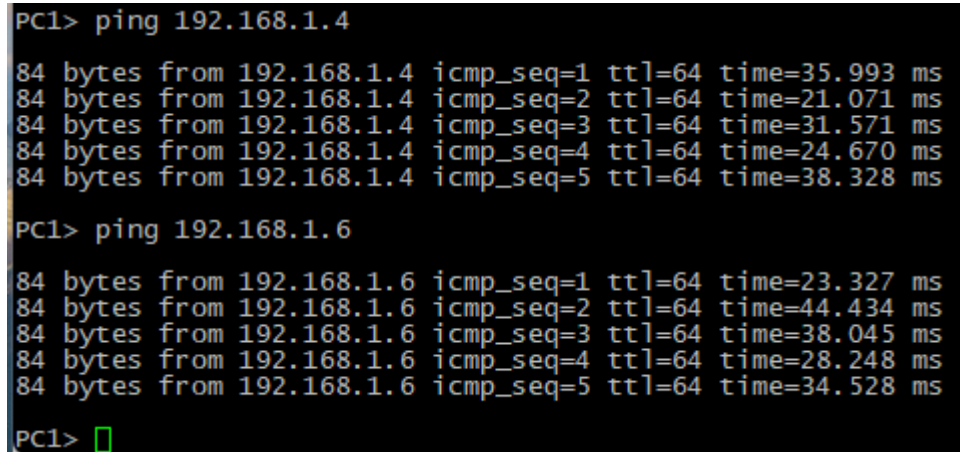
```
enable
configure terminal
interface range GigabitEthernet0/0-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 333
  switchport trunk allowed vlan 20,333
  no shutdown
exit

interface GigabitEthernet1/0
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface GigabitEthernet1/1
  switchport mode access
  switchport access vlan 333
  no shutdown
exit
```

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать.

VLAN 20:



```
PC1> ping 192.168.1.4
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=35.993 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=21.071 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=31.571 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=24.670 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=38.328 ms

PC1> ping 192.168.1.6
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=23.327 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=44.434 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=38.045 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=28.248 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=34.528 ms

PC1> █
```

VLAN 333:

```
PC2> ping 192.168.1.5
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=39.793 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=33.117 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=30.516 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=54.998 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=48.146 ms

PC2> ping 192.168.1.7
84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=76.300 ms
84 bytes from 192.168.1.7 icmp_seq=2 ttl=64 time=28.778 ms
84 bytes from 192.168.1.7 icmp_seq=3 ttl=64 time=19.187 ms
84 bytes from 192.168.1.7 icmp_seq=4 ttl=64 time=39.270 ms
84 bytes from 192.168.1.7 icmp_seq=5 ttl=64 time=55.894 ms

PC2> █
```

Недоступно из разных сетей:

```
PC1 - PuTTY
PC1> ping 192.168.1.3
host (192.168.1.3) not reachable
PC1> ping 192.168.1.5
host (192.168.1.5) not reachable
PC1> █
```

```
PC2 - PuTTY
PC2> ping 192.168.1.4
host (192.168.1.4) not reachable
PC2> ping 192.168.1.6
host (192.168.1.6) not reachable
PC2> █
```

PC, находящиеся в пределах одного VLAN, доступны друг другу. Узлы VLAN20 успешно обмениваются ICMP-пакетами между собой, аналогично у VLAN333. При попытке обмена данными между узлами, принадлежащими различным VLAN, ICMP Echo Reply не получен.

3) Перехватить в Wireshark пакеты с тегами и без тегов (nb!), результаты задокументировать.

С тегами:

*Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

vlan

No.	Time	Source	Destination	Protocol	Length	Info
25	31.154258	Private_66:68:02	Private_66:68:00	ARP	68	192.168.1.4 is at 00:50:79:66:68:02
26	31.168401	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x7b69, seq=1/256, ttl=64 (reply in 27)
27	31.205345	192.168.1.4	192.168.1.2	ICMP	102	Echo (ping) reply id=0x7b69, seq=1/256, ttl=64 (request in 26)
28	32.223317	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x7c69, seq=2/512, ttl=64 (reply in 29)
29	32.242339	192.168.1.4	192.168.1.2	ICMP	102	Echo (ping) reply id=0x7c69, seq=2/512, ttl=64 (request in 28)
30	32.474128	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
31	33.143599	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
32	33.259822	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x7d69, seq=3/768, ttl=64 (reply in 33)
33	33.275762	192.168.1.4	192.168.1.2	ICMP	102	Echo (ping) reply id=0x7d69, seq=3/768, ttl=64 (request in 32)
34	34.293873	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x7f69, seq=4/1024, ttl=64 (reply in 35)
35	34.307764	192.168.1.4	192.168.1.2	ICMP	102	Echo (ping) reply id=0x7f69, seq=4/1024, ttl=64 (request in 34)
36	35.328839	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x8069, seq=5/1280, ttl=64 (reply in 37)
37	35.342250	192.168.1.4	192.168.1.2	ICMP	102	Echo (ping) reply id=0x8069, seq=5/1280, ttl=64 (request in 36)
39	36.003575	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
40	36.780668	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:48:71:08:00:00 Cost = 0 Port = 0x8001

Frame 32: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)

802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20

Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.4

Internet Control Message Protocol

Без тегов:

*Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
2	0.828382	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
3	4.446787	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
4	4.196133	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
5	4.323950	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.5? Tell 192.168.1.3
6	4.337833	Private_66:68:03	Private_66:68:01	ARP	64	192.168.1.5 is at 00:50:79:66:68:03
7	4.352383	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x6f6c, seq=1/256, ttl=64 (reply in 8)
8	4.385497	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x6f6c, seq=1/256, ttl=64 (request in 7)
9	5.391499	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x706c, seq=2/512, ttl=64 (reply in 10)
10	5.418888	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x706c, seq=2/512, ttl=64 (request in 9)
11	6.433420	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x716c, seq=3/768, ttl=64 (reply in 12)
12	6.455515	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x716c, seq=3/768, ttl=64 (request in 11)
13	7.038957	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
14	7.460209	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x726c, seq=4/1024, ttl=64 (reply in 15)
15	7.485354	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x726c, seq=4/1024, ttl=64 (request in 14)

Frame 9: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on 0

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:03 (00:50:79:66:68:03)

Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.5

Internet Control Message Protocol

При передаче трафика VLAN20 в Ethernet-кадре присутствует заголовок IEEE 802.1Q. При передаче трафика VLAN333 заголовок IEEE 802.1Q в кадре отсутствует.